

Comprehensive Summary of Score-Based Generative Modeling through Stochastic Differential Equations

Forward Diffusion Process

The data distribution p_0 is perturbed into a noise distribution p_T via a forward SDE:

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w},$$

where $\mathbf{w}$ is a Wiener process. Three SDE variants are highlighted:

1. **Variance Exploding (VE) SDE:**

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}}d\mathbf{w}, \quad \text{with } \sigma(t) = \sigma_{\min} \left(\frac{\sigma_{\max}}{\sigma_{\min}} \right)^t.$$

2. **Variance Preserving (VP) SDE:**

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}dt + \sqrt{\beta(t)}d\mathbf{w}, \quad \beta(t) = \beta_{\min} + t(\beta_{\max} - \beta_{\min}).$$

3. **sub-VP SDE** (modified VP with bounded variance):

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}dt + \sqrt{\beta(t) \left(1 - e^{-2 \int_0^t \beta(s)ds} \right)}d\mathbf{w}.$$

Perturbation Kernels (Gaussian Transitions)

- For VE:

$$p_{0t}(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}(\mathbf{x}(t); \mathbf{x}(0), \sigma(t)^2 \mathbf{I}).$$

- For VP:

$$p_{0t}(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N} \left(\mathbf{x}(t); \mathbf{x}(0) e^{-\frac{1}{2} \int_0^t \beta(s)ds}, \mathbf{I} \left(1 - e^{-\int_0^t \beta(s)ds} \right) \right).$$

Reverse Generation Process

The reverse-time SDE generates data by evolving noise back to the data distribution:

$$d\mathbf{x} = [\mathbf{f}(\mathbf{x}, t) - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x})] dt + g(t) d\bar{\mathbf{w}},$$

where $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ is the **score**, approximated by a neural network $\mathbf{s}_{\theta}(\mathbf{x}, t)$.

Training Objective

Train $\mathbf{s}_{\theta}(\mathbf{x}, t)$ to match the score of the perturbed data distribution:

$$\theta^* = \arg \min_{\theta} \mathbb{E}_t \left[\lambda(t) \mathbb{E}_{\mathbf{x}(0), \mathbf{x}(t)} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log p_{0t}(\mathbf{x}(t) | \mathbf{x}(0)) \right\|_2^2 \right],$$

where $\lambda(t)$ weights different noise levels (e.g., $\lambda(t) \propto 1/\mathbb{E}[\|\nabla \log p_{0t}\|^2]$).

Sampling Methods

1. **Numerical SDE Solvers** (e.g., Euler-Maruyama):

$$\mathbf{x}_i = \mathbf{x}_{i+1} - [\mathbf{f}(\mathbf{x}_{i+1}, t_{i+1}) - g(t_{i+1})^2 \mathbf{s}_{\theta}(\mathbf{x}_{i+1}, t_{i+1})] \Delta t + g(t_{i+1}) \sqrt{\Delta t} \mathbf{z}.$$

2. **Predictor-Corrector (PC) Samplers:**

- **Predictor:** SDE solver step (e.g., reverse diffusion).
- **Corrector:** MCMC step (e.g., Langevin dynamics) to refine samples.

3. **Probability Flow ODE** (deterministic):

$$d\mathbf{x} = \left[\mathbf{f}(\mathbf{x}, t) - \frac{1}{2} g(t)^2 \mathbf{s}_{\theta}(\mathbf{x}, t) \right] dt.$$

Enables fast sampling via ODE solvers (e.g., RK45) and exact likelihood computation.

Architecture Improvements

- **NCSN++** and **DDPM++**: Enhanced architectures with:
 - Anti-aliasing filters (FIR).
 - Rescaled skip connections ($\times 1/\sqrt{2}$).
 - BigGAN-style residual blocks.
 - Progressive growing for high-resolution images (e.g., 1024×1024).

Controllable Generation

Solve conditional reverse-time SDEs for tasks like inpainting or class-conditional generation:

$$d\mathbf{x} = [\mathbf{f}(\mathbf{x}, t) - g(t)^2 (\nabla_{\mathbf{x}} \log p_t(\mathbf{x}) + \nabla_{\mathbf{x}} \log p_t(\mathbf{y}|\mathbf{x}))] dt + g(t) d\bar{\mathbf{w}},$$

where $\nabla_{\mathbf{x}} \log p_t(\mathbf{y}|\mathbf{x})$ is obtained from a classifier or heuristics.

Key Equations for Implementation

1. **Forward Perturbation** (VE example):

$$\mathbf{x}(t) = \mathbf{x}(0) + \sigma(t)\mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(0, \mathbf{I}).$$

2. **Score Matching Loss**:

$$\mathcal{L} = \mathbb{E}_{t, \mathbf{x}(0), \mathbf{z}} \left[\lambda(t) \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) + \frac{\mathbf{z}}{\sigma(t)} \right\|^2 \right].$$

3. **Reverse Sampling** (VP example):

$$\mathbf{x}_{i-1} = \frac{1}{\sqrt{1 - \beta_i}} (\mathbf{x}_i + \beta_i \mathbf{s}_{\theta}(\mathbf{x}_i, i)) + \sqrt{\beta_i} \mathbf{z}_i.$$

4. **Probability Flow ODE** (deterministic):

$$\mathbf{x}_{i-1} = \mathbf{x}_i + \left[-\frac{1}{2} \beta_i \mathbf{x}_i - \frac{1}{2} \beta_i \mathbf{s}_{\theta}(\mathbf{x}_i, i) \right] \Delta t.$$

Applications

- **Image Generation**: Achieved SOTA FID (2.20) on CIFAR-10.
- **Inverse Problems**: Image inpainting, colorization, and class-conditional sampling via conditional SDEs.
- **High-Resolution Synthesis**: First 1024×1024 samples from score-based models.

Implementation Steps

1. Choose an SDE (e.g., VE/VP).
2. Train a score network $\mathbf{s}_{\theta}$ using the perturbed data and loss.
3. Sample using reverse SDE/ODE solvers.
4. For conditional tasks, integrate task-specific gradients (e.g., $\nabla \log p(\mathbf{y}|\mathbf{x})$).